

TERRA TRACE

FORENSIC AUDIT ENCLAVE & DECISION-SUPPORT SYSTEM

Official Operational User Manual & Team Defense Playbook

SYSTEM DESIGNATION: Air-Gapped Forensic Decision Register
LEGAL FRAMEWORK: Section 65B, Indian Evidence Act (Certificate of Electronic Records)
CORE CONSTITUTIONAL RULE: Non-punitive decision support. Zero automated disqualifications. Every statistical anomaly is delivered to human judicial/administrative officers for statutory adjudication.

1. The 5-Minute Winning Pitch (Presenter Script)

THE 30-SECOND ELEVATOR HOOK:
"Every exam integrity scandal in India—from WBSSC to NEET—takes months to investigate because physical OMR sheets, server databases, and seating plans live in separate departmental silos. **Terra Trace unites all three** into an air-gapped forensic decision register that proves fraud in seconds with immutable Section 65B mathematical evidence."

The 5-Step Live Demonstration Click Path:

- 1. Ingestion Register (Tab 01):** Click Load WBSSC Case (Real). Point out the SHA-256 block hashes: 'Before any math runs, input datasets are cryptographically fingerprinted to satisfy Section 65B court chain of custody.'
- 2. Raw Forensic Inspector (Tab 01):** Click [Watch Raw Datasets]. Point to red highlighted cells: 'Candidate WBSSC_SLST_0001 scored 4.0 on paper OMR, but server database published 53.0. An unauthorized +49 marks inflation.'
- 3. Triangulation Engine (Tab 02):** Explain the 45% Reconciliation / 35% Macro / 20% Micro weights. 'Deterministic mathematical forensics—zero hallucinating AI.'
- 4. Triage Queue & Drilldown (Tab 03):** Open candidate WBSSC_SLST_0001. Walk through the 60-question response grid, the Bell Curve, and the 5x5 Seating Proximity grid.
- 5. Court Dossier & Adjudication:** Click Export PDF Dossier to show the formal Section 65B Certificate. Click Adjudicate to record a confirmed determination with mandatory logged justification.

2. The 3-Layer Forensic Triangulation Model

$$\text{CombinedRisk} = (0.45 \times \text{Reconciliation}) + (0.35 \times \text{Macro}) + (0.20 \times \text{Micro})$$

Layer	Weight	Mathematical Metric	Evidentiary Trigger Condition
Layer 1: Reconciliation	45%	Exact Item Discrepancy: $\Delta = \text{Score_Server} - \text{Score_OMR}$	$\Delta \geq +10.0$ marks flags high-risk database tampering. Distinguishes small answer key updates from manual mark inflation.
Layer 2: Macro Centre	35%	Two-Sample Kolmogorov-Smirnov (KS) Test & Skewness	$D > 0.35$ and $p < 0.01$ indicates centre-level paper leak or localized broadcast resulting in anomalous score clustering.

Layer 3: Micro Seating	20%	Adapted Wollack ω Index & Holland K-Index	Adjacent desks ($d \leq \sqrt{2}$) sharing ≥ 6 identical incorrect answers ($z > 3\sigma$) flags organized spatial seating collusion.
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3. The 5 Core System Registers

Register	Name & Purpose	Core Capabilities
Tab 01	Ingestion & Checkpoints Schema gatekeeper	Strict 7-column schema verification, SHA-256 block hashing, 3 scenario presets (WBSSC Real, High-Risk Anomaly, Clean Control), and Raw Dataset Forensic Inspector.
Tab 02	Analytical Engine Forensic calculation	3-Layer triangulation pipeline, composite scoring weights, and Computer-Based Test (CBT) roadmap (mouse dynamics, response latency, LAN socket jitter).
Tab 03	Triage Queue & Adjudication Investigative workspace	Multi-column triage filtering, 3-layer candidate drilldown (OMR audit, Bell curve, 5x5 seating grid), Section 65B PDF dossier export, and human adjudication.
Tab 04	Methodology & Standards Statutory compliance	Documentation on Indian Evidence Act admissibility, error rate controls (Benjamini-Hochberg FWER < 0.001), and non-punitive decision support principles.
Tab 05	Cryptographic Audit Trail Chain of custody	Tamper-evident, append-only SQLite log recording file byte sizes, SHA-256 fingerprints, timestamps, and officer badge IDs.

4. Cloud Deployment & Network Architecture

Frontend (Vercel): React 19 single-page application. Deployed at your Vercel project URL.

Backend (Render): Python FastAPI web service with SQLite and Polars analytics. Check health at: <https://<your-render-name>.onrender.com/health>.

Understanding Render Free-Tier Sleep: Inactive instances sleep after 15 minutes. The initial wake-up call takes **30–45 seconds**. All subsequent queries execute in **under 100ms**. *Pro Tip: Visit the health URL once before presenting.*

Header Connection Pill: Click [API LIVE / SET BACKEND URL] in the top right header to test server latency live or switch backend endpoints.

Offline Fail-Safe: If internet is unavailable, Terra Trace automatically runs the embedded offline enclave so your demo never freezes or fails.

5. Defense Q&A; for Judges & Stakeholders

Q1: Is this using Artificial Intelligence or Machine Learning?

"No, and that is our strongest feature. In a court of law, 'black box' machine learning weights and LLM hallucinations are legally inadmissible. Terra Trace uses deterministic statistical forensics: exact item reconciliation, two-sample KS distribution tests, and Wollack ω seating proximity indices. Every score is explainable down to the exact math."

Q2: Can an auditor tamper with evidence to frame a student?

"No. Terra Trace implements an append-only cryptographic audit register. The instant raw files are uploaded, their

SHA-256 block hashes are stamped immutably. Any manual change to an OMR CSV file alters the cryptographic hash, immediately invalidating the Section 65B legal certificate."

Q3: What if an entire centre did well because they had great teachers?

"The Macro layer does not flag centres for high average marks. It tests for distribution anomalies—such as an unnatural cluster strictly at the qualifying cutoff (50-55 marks) with zero room variance. Normal high-performing schools exhibit natural Gaussian bell curves across their student cohort."

Q4: How do you protect student privacy under the DPDP Act?

"Terra Trace is designed to run air-gapped on on-premise government servers without internet access. Candidate names, phone numbers, and Aadhaar numbers are never ingested—only anonymized Roll IDs and Seat Coordinates."

6. Quick Troubleshooting Cheat Sheet

Symptom	Cause	Immediate Fix
Cards spinning on 'Hashing in Progress'	Render free instance is booting up from sleep.	Open https://<render-url>/health in a tab to wake it up. Or press Ctrl+F5 to run embedded enclave.
Mobile Chrome shows old cached screen	Browser cached previous JavaScript bundle.	Open in an Incognito Tab (Ctrl+Shift+N), or clear cached images and files in Chrome settings.
Export PDF Dossier opens print dialog	Offline or mobile browser environment.	Select 'Save as PDF' directly in the native mobile/desktop print dialog.